

SURFACE FEATURES

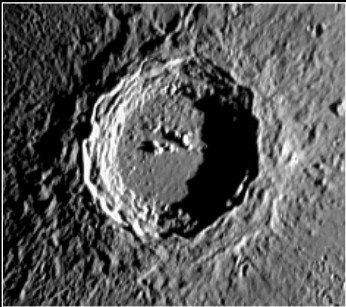
You can train your eye to observe detail by concentrating on small parts of the lunar surface and sketching the craters. You will soon realize that there are far more small craters than large ones. The biggest visible crater is Mare Imbrium, some 680 miles (1,100 km) across, the formation of which almost broke the Moon apart. About 4 percent of the craters are not circular. These were produced by impactors arriving at very low angles. Some craters are “ghosts”—only the peaks of the walls protrude above the mare lava. The lunar surface also has some interesting valleys. These were not caused by flowing water but are mainly the remnants of lava tubes that have emptied and then caved in. Smaller valleys are called rills. Craters such as Gassendi and Hevelius have many rills criss-crossing their lava-filled basins. These were formed when the lava cooled and contracted.



Terminator
The terminator is the boundary between the sunlit part of the Moon and the dark part. It moves around the Moon during the month. Shadows at the terminator are very long, and features such as mountains are thrown into relief.



NOON: THE SUN IS OVERHEAD



SUNRISE: THE SUN IS LOW

Sunrise over Copernicus
This crater is 800 million years old, and is 57 miles (91 km) wide and 2.3 miles (3.7 km) deep. It looks “washed out” when the Sun is overhead (top), but at sunrise (above), the long shadows highlight the dramatic nature of the crater’s collapsed inner walls and the central mountain. The surrounding hills are also more visible.



Sinus Iridum
The beautiful Sinus Iridum (Bay of Rainbows) is the western half of a 160-mile- (260 km-) diameter impact crater that tilted over when the central regions of Mare Imbrium subsided. The eastern side of the crater was then lost from view as lava from beneath the lunar crust completely flooded the gigantic mare.



Tycho Crater
Visible in the center of this image, the Tycho Crater is an impressive sight through binoculars. This “young,” 52-mile- (85-km-) wide crater was formed 100 million years ago. In its center is a 1.8-mile- (3-km-) high mountain peak, formed when the underlying rock relaxed after being depressed by the pressure of the impact. Bright rays of material that were ejected from the crater can be seen splashed over the nearby lunar surface.

OBSERVING CRATERS

Craters appear at their most dramatic when they are near the terminator, which sweeps across the lunar surface as the lunar month progresses. The craters listed here have been divided into four groups according to when they are close to the terminator—that is, whether they are best observed when the Moon is a new waxing crescent, at first quarter, at last quarter, or an old waning crescent. The craters can all be viewed with binoculars. When the Moon is full, concentrate on the cratered regions around its rim.

BEST OBSERVATION TIME

Waxing crescent	Last quarter
First quarter	Waning crescent
Endymion	Plato
Langrenus	Tycho
Petavius	Eratosthenes
Atlas	Aristarchus
Albategnius	Gassendi
Ptolemaeus	Grimaldi
Aristillus	Pythagoras

LUNAR ECLIPSES

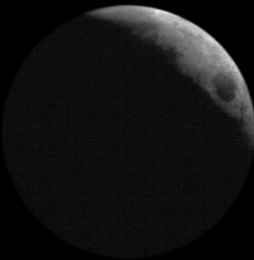
Total lunar eclipses occur only when the Moon is full and the Sun, Earth, and Moon are lined up (see p.15). Such eclipses can be seen from anywhere on the nighttime hemisphere of Earth, and no special equipment is needed to observe one. During the eclipse, the Moon is in the Earth’s shadow, and the Earth is preventing sunlight from reaching its surface. At totality, a small amount of indirect sunlight still manages to reach the Moon. This travels close to the Earth and through its atmosphere. As a result, the light is a deep red-orange color.



Onset of eclipse
The lower edge is totally eclipsed. Light from part of the Sun continues to shine on the rest of the Moon.



Halfway stage
About half of the Moon is now totally eclipsed, while the other half is still illuminated by sunlight.



Darkened crescent
Most of the Moon is in Earth’s shadow. The crescent is faint because it is in the penumbra.



Totality
The Moon is completely inside Earth’s shadow and is a dusky reddish color.